

GET EMBEDDED RIGHT AWAY!

Right from the chip designing to system architecture and formulating testing strategy for software, embedded systems open up a plethora of opportunities for their practitioners

■ SUDESHNA DAS

Career in embedded systems demands a discussion in continuation with my article last month on the vocational challenges in the microcontroller industry. Embedded systems are the platform for the application of microcontrollers. So I put everything away, dug deep into these and came out with a laundry list that should never, and I mean never, be neglected if you want to be in this sector. Here's your what-will-get-you-hired guide.

What's the market like?

Embedded systems are omnipresent. These are there in your home, college, office, shopping mall and so on and so forth. Even while you are on the move—either on a most basic two-wheeler or on an advanced aircraft—you are still amidst embedded systems. They are a unique combination of computer hardware, software and perhaps additional mechanical or other parts, designed to perform a specific function within a given time frame.

The embedded software is required for all real-time applications and developed using a real-time operating system (RTOS), as it helps to schedule and execute tasks based on the priority in a predictable manner. To cite just a few examples, embedded software allow your washing machine to choose speed according to the type of cloth, confer thinking power to the microwave ovens and propel rocket launchers into space.

Industry analysts estimate that

Our campus recruitment is based on a written test and an interview. The written test is separate for hardware and software engineers. The hardware paper has a section on aptitude, a section on digital design and a section on analogue design—we ask the students to select any one section between digital and analogue design. The software paper has an aptitude section and a section on software. Candidates who fare well in the written test are shortlisted for interview. From my experience, the interviews are mostly based on problem-solving exercises and test the conceptual understanding rather than rote learning.



—C.P. Ravikumar, technical director, university relations, Texas Instruments

over 54 per cent of future software development will be in the embedded technology space. According to an IDC report, the international market as a whole expects product development worth \$75 billion.

The Indian scenario

In spite of job cuts due to recession, semiconductor companies in India are in hiring mode. India is considered one of the emerging markets in the semiconductor space. According to an ISA-IDC report on the Indian semicon-

ductor and embedded design service industry (2007-10), the Indian semiconductor design market is expected to cross \$7.37 billion (Rs 317 million) in 2008. The Indian design market is pegged to grow at a compounded annual growth rate (CAGR) of around 21.7 per cent between 2007 and 2010.

This tremendous rate of growth will require a large number of skilled professionals. The semiconductor design services industry in India employed an estimated 150,000 people in 2007-08. Of this, the bulk of the jobs